

metrics.

Planning parameters. Regardless of the features used for planning, the performance of VAE-IW depends on the variant of RolloutIW being used, and on its parameters. In Table ?? we compare the average score of VAE-IW(1) with and without risk aversion and observe that the risk-averse variant achieves an equal or higher average score in 68% of the domains. Table ?? shows the results of VAE-IW(1) and RolloutIW(1) with B-PROST features, similarly to Table 1, except that both methods are run *without* risk aversion. With this modification, our method still obtains a higher average score in the majority of games (68%).

Another crucial planning parameter is the time budget for planning at each time step. While the main results are based on a 0.5s budget, we also consider a 32s budget, following ?. In Table ?? we observe that, unsurprisingly, the 32s budget leads to better results in most domains (79%). Interestingly, increasing the planning budget does not seem to affect the average rollout depth, while the average number of expanded nodes in each planning phase grows significantly. This behaviour is consistently observed in all tested domains (see Figures ?? and ?? in Appendix) and points to the fact that increasing the planning budget improves results mostly by allowing more rollouts.

In Table ?? we compare the average scores obtained by VAE-IW(1) and RolloutIW(1) with B-PROST features, using a 32s planning budget. Once again, using the compact features learned by a VAE seems to be beneficial, as VAE-IW(1) obtains the highest average score in 62% of the games. However, the margin here is smaller than in the experiments with a budget of 0.5s, where VAE-IW(1) outperformed RolloutIW(1) B-PROST in 73% of the games.

Related Work

Variational Autoencoders (VAEs) have been extensively used for representation learning (?) as their amortized inference network lends itself naturally to this task. In the context of automated planning, ? proposed a State Autoencoder (SAE) for propositional symbolic grounding. SAEs are in fact VAEs with Bernoulli latent variables. They are trained by maximizing a modified ELBO with an additional entropy regularizer, defined as twice the negative KL divergence. Thus, the objective function being maximized is the ELBO (1) with the sign of the KL flipped. A variation of SAE, the zero-suppressed state autoencoder (?), adds a further regularization term to the propositional representation, leading to more stable features.

? learn state features with supervised learning, and then plan in the feature space with a learned transition graph. ? specify a set of skills for a task, and automatically extract state representations from raw observations. ? use generative adversarial networks to learn structured representations of images and a deterministic dynamics model, and plan with graph search. ? proposed π -IW, a variant of RolloutIW where a neural network guides the action selection process in the rollouts, which would otherwise be random. More-

over, π -IW plans using features obtained from the last hidden layer of the policy network, instead of B-PROST.

Atari games have also been extensively used as a testbed in model-free (?????) and model-based (????) deep reinforcement learning.

Conclusions

We have introduced a novel combination of width-based planning with learning techniques. The learning employs a VAE to learn relevant features in Atari games given images of screen states as training data. The planning is done with RolloutIW(1) using the features learned by the VAE. Our approach reduces the size of the feature set from the 20.5 million B-PROST features used in previous work in connection with RolloutIW, to only 4,500. Our algorithm, VAE-IW, outperforms the previous methods, proving that VAEs can learn meaningful representations that can be effectively used for planning with RolloutIW.

In VAE-IW, the symbolic representations are learned from data collected by RolloutIW using B-PROST features. Increasing the diversity and quality of the training data could potentially lead to better representations, and thus better planning results. One possible way to achieve this could be to iteratively retrain and fine-tune the VAEs based on data collected by the current iteration of VAE-IW. The quality of the representations could also be improved by using more expressive discrete models, for example with a hierarchy of discrete latent variables (?). Similarly, in the ablation study where we compare this approach with quantizing continuous representations, more flexible models (??) and more advanced quantization strategies could be employed. Finally, we can expect further improvements orthogonal to this work, by learning a rollout policy for more effective action selection as in ?.

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